Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/31/2006
PI Name:	Thomas, James David M.D.		
Project Title:	Diagnostic Three Dimensional Ultrasonography: Development of Novel Compression, Segmentation and Registration Techniques for Manned Space Flight Applications		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Smart Medical Systems Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	thomasj@ccf.org	**Fax:**	FY 216-445-7306
PI Organization Type:	NON-PROFIT	**Phone:**	216-445-6312
Organization Name:	The Cleveland Clinic Foundation		
PI Address 1:	Cardiovascular Medicine		
PI Address 2:	9500 Euclid Ave		
PI Web Page:			
City:	Cleveland	**State:**	OH
Zip Code:	44195-0001	**Congressional District:**	11
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	07/01/2001	**End Date:**	10/31/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-SMS.002.06		
Performance Goal No.:			
Performance Goal Text:			
	The National Space Biomedical Research Institute has identified that the efficient and automated delivery of health care in space is a key research arena for the future. Specifically, they propose to develop a “Smart Medical Care System" that will be able to monitor crew health, identify deviations from ground-based norms, and allow timely intervention by crew members who may have only a moderate amount of training in medicine. Such a Smart Medical Care System will have multiple components, including sensors, data analysis, data integration, and automated effectors that will ultimately deliver care. Although the development of novel sensor technology will undoubtedly be pursued in this program, it is clear that for the foreseeable future, the principal diagnostic imaging modality for crew use in space will be ultrasound. Therefore, a key goal of the Smart Medical Care System should be to optimize the use of diagnostic ultrasound in space and integrate it in an intelligent fashion with decision-making software. For the last three years, the principal investigator and colleagues have worked closely with NASA scientists, flight surgeons, and engineers to optimize		

Task Description:	research and diagnostic ultrasound aboard the International Space Station (ISS) and thus are well positioned to develop the necessary tools and techniques to integrate ultrasound into the Smart Medical Care System. One of the principal limitations of ultrasound technology is its extreme dependence on the expertise of both the acquiring examiner and the interpreting physician. This is particularly true of two-dimensional ultrasound, where the examiner is required to obtain precisely oriented anatomical sections of the organ of interest. Three-dimensional (3D) ultrasound has the advantage of acquiring a large anatomic volume from a single ultrasonic window and thus may be less dependent upon the expertise of the examiner. Furthermore, this large volume may contain sufficient anatomic landmarks to allow unambiguous registration with previously obtained 3D data from either ultrasound or other modalities such as magnetic resonance imaging (MRI) or computed tomography (CT). One could thus envision a system by which whole organs or even the entire body would be imaged in 3D prior to launch, data which could be used to compare with subsequently obtained 3D datasets using in-flight ultrasonography. The overall purpose of this grant is therefore to perform ground-based research, development, and validation aimed at optimizing diagnostic ultrasound in manned spaceflight, with the following unifying hypothesis: Serial 3D ultrasound examinations will enhance diagnostic capabilities in manned spaceflight. This hypothesis will be pursued with both hardware and software development, along with extensive validation studies. Although many disease models could serve for testing the system, we will choose to study two conditions that are highly relevant to problems associated with long-term spaceflight: nephrolithiasis and ventricular atrophy. The technical aspects of this program will be pursued with the following specific aims: 1) Optimize acquisition methods for 3D sonography using reconstruction and real-time techniques. 2) Develop techniques for registering anatomical images from 2D and 3D ultrasound with those obtained from prior ultrasound examinations and from magnetic resonance and computed tomographic imaging, considered “gold standards” for non-invasive anatomical imaging. 3) Develop tools for abstracting, in an automated fashion, anatomic changes from serial 3D and 2D ultrasound studies. 4) Develop algorithms for the optimal compression of 3D ultrasound images. 5) Assess the ability of novice examiners to obtain 3D sonographic datasets after minimal training. These objectives will be pursued using data from a variety of in vitro, animal and clinical models. In particular, we will take advantage of a well-established collaboration with the National Institutes of Health, which permits highly sophisticated chronic animal models to be examined with a minimum of additional resources. Although the tools developed here should be applicable to any organ of the body, we will focus our efforts in two areas that present unique challenges and also are highly relevant to spaceflight. 1) The Kidneys: Bone demineralization during prolonged spaceflight may increase the risk for kidney stones. From a technical perspective, the kidney may be a particularly easy organ to register in three dimensions with prior examinations, being static and having well-defined anatomical landmarks. Using computed tomography as the reference standard, we will examine serially patients with renal and ureteral calculi, paying particular attention to the development and resolution of hydronephrosis and calyceal dilation. 2) The Heart: Manned space flight and prolonged exposure to microgravity may result in significant myocardial atrophy. Furthermore, cardiovascular emergencies such as myocardial infarction and cardiac trauma produce structural changes that are important in guiding therapy. From a technical perspective, the heart is a more difficult organ to register than the kidney, since it is dynamic. The presence of well-defined anatomical landmarks, however, assists in this process. To better understand the process of cardiac remodeling with unloading, we will examine two groups of patients in whom sudden changes in chronic loading conditions are known to produce profound ventricular remodeling: 1) patients with aortic stenosis undergoing aortic valve replacement; 2) patients with chronic severe aortic insufficiency undergoing aortic valve replacement or repair. These patient groups will provide insight into the determinates of ventricular remodeling, while forming an important testbed for the assessment of our registration and comparison algorithms for detecting changes in ventricular morphology over time. In addition to the practical application to manned spaceflight and the ISS, these systems each represent significant societal problems for which the incidence of disease is common enough to study within the time period of this grant. Furthermore, ultrasound is the standard diagnostic tool used in the management of patients with each of these conditions. At the conclusion of this project, we anticipate delivering to the National Space Biomedical Research Institute and its Smart Medical Care System a set of algorithms and software for the nonrigid morphological registration and comparison of serial two and three dimensional ultrasound datasets and validated algorithms for optimal compression of 3D ultrasound data. In addition to these technical deliverables, our validation work on nephrolithiasis will provide important diagnostic clues for assessing this condition in manned spaceflight. Similarly, the work on cardiac mass regression following unloading will be invaluable to the NASA research and medical operations community in assessing the impact of long-term spaceflight on cardiac atrophy and utility of prophylactic countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	• Real-time 3D ultrasound is accurate for cardiac volumes and function • Mitral annular motion useful for assessing LV function • 3D aortic annular motion predictive for global LV function • Wavelet packet compression can be applied to 3D ultrasound at ratios > 100:1 • Multimodality registration successfully accomplished
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 04/09/2019)